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A CWGAN-GP Based Efficient Proxy Model for Predicting Pressure Profiles in Oil Reservoir

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Abstract

Accurate prediction of reservoir pressure can timely obtain corresponding adjustment countermeasures, which is of great significance for scientifically and rationally formulating mining plans and improving recovery rates. Usually, the production dynamics prediction adopts numerical simulation methods, but the complexity of reservoir geological conditions will greatly increase the difficulty and time required to achieve a complete simulation. The rapid development of technologies such as big data and artificial intelligence has provided a new perspective for solving this problem. Based on the theory of CGAN, the scope of application of the model is improved, and six conditional parameters including time, matrix permeability, and fracture permeability are added to achieve real-time dynamic prediction of reservoir pressure under any conditional parameters. The increase in conditional parameters improves the generalization of the model, but also increases the difficulty of convergence of the model. Therefore, the loss function is optimized by combining the Wasserstein distance and gradient penalty theory, forming a reservoir pressure prediction proxy model that is extremely easy to converge and has strong generalization. This model considers the influence of multi-scale reservoir uncertainty and well control changes, and successfully predicts the pressure at any time under fracturing horizontal well production, with an accuracy of more than 98%, and an efficiency that is 2-3 data levels higher than numerical simulation. It can be used to solve computationally intensive tasks such as history matching and production optimization, and also provides new ideas for the construction of big data models for dynamic prediction of unconventional reservoirs.

Introduction

As an unconventional resource, shale gas is widely distributed worldwide. However, shale gas production remains low at the current stage, with small development scales, and the overall industry is still in its early stages. In the process of shale gas development, one of the primary methods for obtaining shale reservoir parameters is to dynamically predict pressure of multi-stage fractured horizontal wells. Therefore, accurately predicting reservoir pressure enables timely adjustment strategies, which are of great significance

for the scientific and reasonable formulation of residual gas extraction plans and improvement of recovery rates. The most common pressure prediction methods fall into three categories: geological modelling, reservoir engineering and numerical simulation. Geological modelling methods typically rely on seismic exploration predictions, which are direct measurement methods. However, these methods often produce inaccurate results, and various factors can interfere with the accuracy of the predictions, making it difficult to obtain precise results. Additionally, these methods require substantial investment costs. Reservoir engineering methods generally use approximate techniques involving mathematical modelling to simplify complex engineering problems and rapidly predict shale gas pressure. However, these methods can overly idealize the situation, resulting in less accurate predictions. Currently, numerical simulation methods are the most commonly used. Based on geological modelling, they utilize fluid mechanics theory to predict pressure changes over time, thereby guiding reservoir development. However, such methods are computationally complex and labor-intensive. Notably, history matching and subsequent production optimization require thousands of numerical simulations to predict reservoir dynamics, resulting in significant time losses^[1–4]. Therefore, establishing a new method for efficiently predicting reservoir pressure dynamics that overcomes the limitations of traditional prediction methods, such as high testing costs, idealized assumptions and high computational costs, is a critical issue that needs to be addressed urgently.

Data mining and artificial intelligence technologies are gradually penetrating various industrial sectors and achieving significant results. This is bringing new opportunities to the traditional oil industry, which is currently undergoing a critical period of digital transformation. Intelligent models can reveal hidden correlations within large sets of known data, establish intricate mapping relationships between inputs and outputs, and accurately predict results. In recent decades, various methods have been studied to get around the problems with these other methods of pressure prediction. At first, the methods used to create models were based on linear (for example, multiple linear regression or Kriging) or non-linear (for example, support vector machines and artificial neural networks (ANN)) regression. These methods were used to establish the mapping relationship between the input and output data of reservoir numerical simulations^[5–6]. However, many traditional proxy model construction methods were originally designed for low-dimensional problems. When the input dimension is high, problems arise due to their shallow structure (i.e. a small number of hidden layers) and fully connected nature (i.e. each neuron is connected to all neurons in the next layer). This renders traditional proxy models unsuitable for approximating the high-dimensional space problems frequently encountered in reservoir engineering. The emergence of deep learning—characterized by deep architectures, localized connectivity (e.g., convolutional kernels), and automated feature abstraction—directly addresses these limitations^[7–9]. However, a disadvantage of establishing a deep learning computational framework is that certain parameters often need to be fixed. This limits the model to performing prediction or optimization tasks only under specified conditions, which is highly restrictive. Current deep learning models, for example, can predict pressure or fluid saturation fields at any given time, but they only consider time series or other specific conditions. This means the number of factors considered is limited, resulting in low model generalization. If results under different conditions are required, a new proxy model must be trained for those specific conditions, which limits its applicability^[10–12].

During the development of oil reservoirs, numerous parameters have a significant impact on pressure distribution within the reservoir. This study proposes a deep learning framework based on the Conditional Wasserstein Generative Adversarial Neural Network with Gradient Penalty (CWGAN-GP) model to address this limitation. The framework's goal is to create a flexible proxy model that can predict pressure distributions in any reservoir conditions. Adjustable parameters include time series, matrix permeability, fracture permeability, compressibility coefficient, initial bottom-hole flow pressure and bottom-hole flow pressure decay coefficient. The primary contribution of this surrogate model is overcoming the narrow applicability limitation of previous deep learning frameworks, thereby enhancing the model's generalizability. The entire framework can also be applied to other prediction tasks, and the corresponding constraint parameters can be appropriately increased or decreased, thereby enhancing the practical

significance of related GAN methods in oil and gas field development. It is worth noting that, under conditions where the training dataset is sufficiently large and the constraints are sufficiently numerous, the deep learning framework proposed in this paper can provide a new approach for establishing large-scale models for reservoir dynamics prediction.

Methodology

Generative Adversarial Network

A Generative Adversarial Network (GAN) is a framework that enables deep learning models to capture the distribution of training datasets, allowing us to generate new data based on this distribution^[13]. It consists of generators and discriminators. The basic theory involves finding the Nash equilibrium point in a high-dimensional data space through mutual competition between the generators and discriminators. The GAN framework diagram is shown in Fig. 1. However, this architecture lacks an explicit condition control mechanism. This structural flaw significantly limits its applicability in scenarios that require the targeted generation of specific attribute data. The optimization objective is solely to make the generated data closely resemble real data in terms of overall distribution. It does not enable targeted control of the generation process based on predefined condition information, such as category labels and attribute characteristics. This non-conditional characteristic results in highly random generation outcomes, which makes ensuring that the generated data is relevant to specific conditions difficult. This non-conditional characteristic results in highly random generation outcomes, which makes ensuring that the generated data is relevant to specific conditions difficult. Secondly, the lack of condition constraints means that the generated distribution does not match the conditional target distribution very well. This results in poor performance in tasks such as multimodal data generation and attribute-controlled generation. Furthermore, it is challenging to utilise supervised label information effectively to enhance the quality and controllability of the generation process, which significantly restricts its applicability in practical reservoir engineering applications that demand precise conditional control.

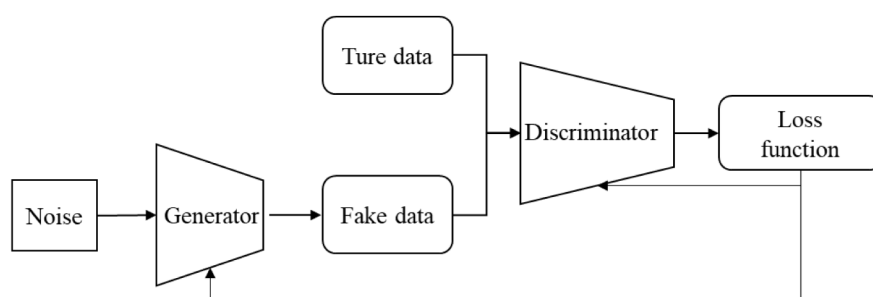


Figure 1—Generative Adversarial Network theoretical diagram

Conditional Generative Adversarial Network

To address these issues, scholars proposed the conditional generative adversarial network (CGAN) model in the same year^[14]. Its structure is shown in Fig. 2. During the establishment of the generator and discriminator, a conditional variable c is introduced. This condition can be any data structure, such as a label, tag or image. The images generated by the generator are more targeted and the discrimination by the discriminator is more focused due to the constraints imposed by the conditions.

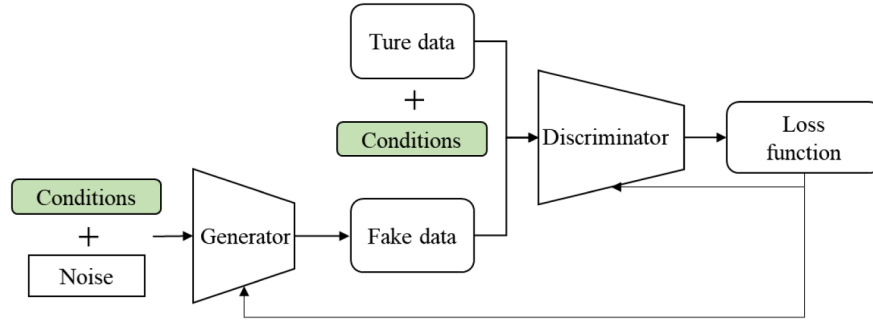


Figure 2—Conditional Generative Adversarial Network theoretical diagram

This conditional mechanism directs the generation process, enabling the synthesis of targeted outputs while enhancing the discriminator's ability to evaluate data authenticity under specific constraints. In the G, the conditional variable c is concatenated with the noise vector z to form a joint latent representation, which is then decoded into synthetic data. In the discriminator, the real data x and conditional variable c are jointly input to assess the likelihood of x being authentic under condition c . Mathematically, the CGAN objective function extends the GAN minimax formulation by conditioning both G and D on c , as shown in Eq. (1):

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}} [\log D(x|c)] + \mathbb{E}_{z \sim p_z, c \sim p_c} [\log(1 - D(G(z|c)|c))] \quad (1)$$

The objective function during training can be formulated as a two-player maximin game involving G and D , whether it is GAN or CGAN, as shown in Equation (1). Unconditional binary cross-entropy is used as the optimization objective and exhibits non-convex and non-smooth characteristics. The generator and discriminator objective functions form a zero-sum game, which can lead to parameter updates that are prone to oscillation or vanishing gradients. Secondly, how conditional variables and noise inputs are combined (e.g. simple concatenation) can dilute conditional information in the feature map. This reduces the sensitivity of the loss function to conditions, leading to ‘conditional collapse’. In other words, the greater the amount of conditional information, the smaller the convergence basin of the loss function becomes, making it more sensitive to hyperparameters such as the learning rate and the iteration strategy.

Wasserstein Generation Adversarial Network with Gradient Penalty

The root causes of the original GAN problem can be summarized as follows: Firstly, the distance measures used for equivalent optimization (e.g. KL divergence and JS divergence) are unreasonable. Secondly, after random initialization, it is difficult to generate a distribution by the generator that significantly overlaps with the real distribution. WGAN addresses the first root cause by replacing the JS divergence with the Wasserstein distance^[15], thereby resolving both training stability and progress metric issues. The Wasserstein distance, also known as the Earth-Mover (EM) distance, is formally defined as:

$$W(P_r, P_g) = \inf_{\gamma \in \Pi(P_r, P_g)} \mathbb{E}_{(x,y) \sim \gamma} \|x - y\| \quad (2)$$

where $\Pi(P_r, P_g)$ denotes the set of all joint $\gamma(x,y)$ whose marginals are P_r and P_g , respectively. Intuitively the Wasserstein distance measures the minimum "cost" of transforming the generated distribution p_{data} into the real distribution p_g , where the cost is quantified by the expected Euclidean distance $\|x - y\|$. $\mathbb{E}_{(x,y) \sim \gamma} [\|x - y\|]$ denotes the expected value of the sample to distance under joint distribution, $\inf$ denotes the lower bound of certainty. The Wasserstein distance is defined as the infimum of the expected value over all possible joint distributions. Its superiority over the KL and JS divergences lies in its ability to quantify the distance between two distributions even when they are disjoint. In high-dimensional spaces, if two different distributions do not overlap or their overlap is negligible, the KL and JS divergences fail to reflect their proximity or provide

meaningful gradients. In contrast, the Wasserstein distance ensures non-vanishing gradients, enabling stable optimization even in non-overlapping regimes.

In summary, Wasserstein GAN incorporate the Wasserstein distance into their loss function. Compared to KL divergence and JS divergence, the Wasserstein metric exhibits superior smoothness properties, theoretically resolving training instability while providing a reliable quantitative indicator for training progression. However, empirical studies reveal persistent challenges in practical implementations, including training sensitivity to hyperparameters and suboptimal convergence rates, which persist despite the theoretical guarantees.

Therefore, scholars proposed the new WGAN-GP model^[16]. The most significant optimization is changing the implementation of Lipschitz constraints from weight clipping to gradient penalty. There are two serious issues with the implementation of weight clipping. The discriminator loss function aims to maximin the difference in scores between real and fake samples. However, weight clipping independently limits the value range of each network parameter to $[-0.01, 0.01]$. The optimal strategy for the loss function in this case is therefore to push all parameters to their extremes, i.e. to take either the maximum value (e.g. 0.01) or the minimum value (e.g. -0.01). This makes the discriminator highly likely to learn a simple mapping function. Secondly, weight clipping can easily lead to gradient vanishing or explosion. This is because the discriminator is a multi-layer network. If the clipping threshold is set slightly too small, the gradient decreases slightly with each layer, leading to exponential decay after multiple layers. Conversely, if it is set slightly too large, the gradient increases slightly with each layer, leading to exponential explosion after multiple layers. Only by setting the threshold precisely can the generator obtain good backpropagation gradients. However, in practical applications, this balance zone may be very narrow and tuning the parameters increases the workload. But using the gradient penalty method to implement Lipschitz constraints effectively resolves both above issues (as shown in Fig. 3).

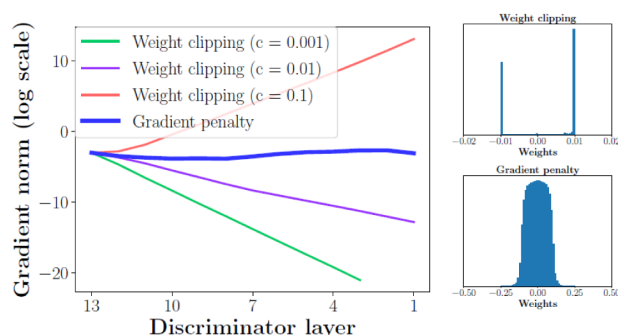


Figure 3—Experimental comparison results of gradient penalty and weight clipping^[16]

Conditional Wasserstein Generation Adversarial Network with Gradient Penalty

In summary, WGAN-GP can theoretically solve the problem of unstable training in CGAN. This paper therefore combines the principle of adding constraints to the original GAN in CGAN with the optimization theory of WGAN-GP for the original GAN loss function, forming a generalized, easily convergent deep learning model. This study aims to identify the potential relationship between crack morphology and pressure distribution data, establishing a dynamic mapping relationship between the two using deep learning methods. Consequently, the generator's input must be high-dimensional feature tensors obtained by encoding fracture morphology, rather than randomly generated noise data. Fig. 4 illustrates the overall structure of the CWGAN-GP framework, consisting of a generator, a critic and a loss function. The generator comprises an encoder and a decoder. The encoder extracts high-dimensional features from input fracture morphology images via convolution calculations, and the decoder restores predicted reservoir pressure distribution images via inverse convolution calculations. The position in which constraints are inserted is determined by their type. Specifically, it depends on whether they are overlaid on the input data or on the

high-dimensional feature tensor obtained through the encoded structure. The critic also needs to overlay the constraints on the input image data in the same way. Note that, in the original GAN, the Discriminator performs a binary classification task, whereas in this architecture, the Critic fits the Wasserstein distance, which is a regression task. Therefore, the sigmoid function in the last layer must be removed. Finally, the loss function of CWGAN-GP is obtained by introducing the constraint c into the loss function of WGAN-GP. The loss functions for the generator and critic are as follows:

$$L(D) = \mathbb{E}_{x \sim P_g}[D(x|c)] - \mathbb{E}_{x \sim P_r}[D(x|c)] + \lambda \mathbb{E}_{x \sim \mathcal{X}} [\|\nabla_x D(x|c)\|_2 - 1]^2 \quad (3)$$

$$L(G) = -\mathbb{E}_{x \sim P_g}[D(x|c)] \quad (4)$$

Among them, c represents the set of constraints corresponding to sample x .

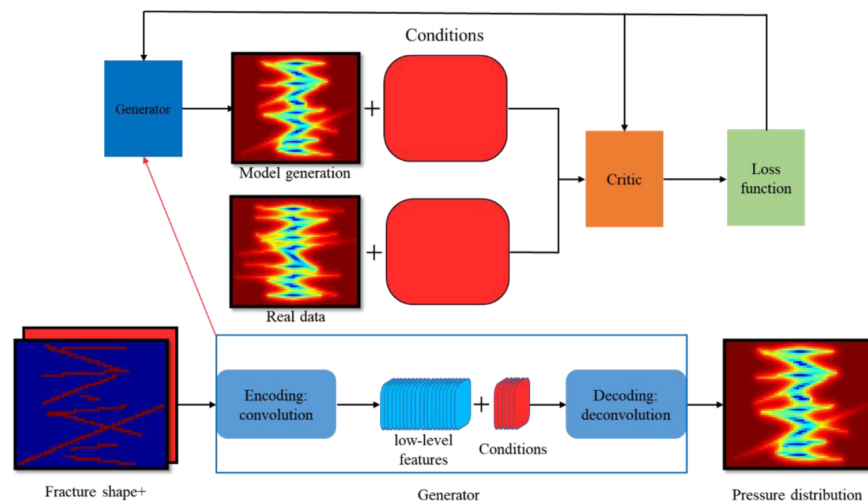


Figure 4—Network structure of CWGAN-GP

Experiment and Results Analysis

Reservoir and dataset description

As is well-established in deep learning practice, training robust models necessitates extensive datasets. Given our objective to develop a generalized surrogate model with broad applicability, this requires substantial reservoir pressure distribution data spanning diverse geological configurations and production scenarios. However, real-world reservoir data often suffers from critical limitations: scarcity of field measurements, data incompleteness/inaccuracies, and insufficient coverage of parameter space. To overcome these challenges, we employ physics-based numerical simulations to systematically generate high-fidelity training datasets that comprehensively capture. As previously delineated, the Embedded Discrete Fracture Model (EDFM) employs structured grid discretization for matrix systems while embedding fracture networks into matrix cells to address multiphase flow in complex fracture systems, thereby enhancing preprocessing efficiency. However, conventional commercial simulators (e.g., Eclipse, CMG) exhibit limitations in effectively handling fracture-matrix misalignment scenarios using simplistic Cartesian grids. To overcome this constraint, we developed an in-house EDFM simulator to generate comprehensive training datasets. In this study, five parameters are defined to represent the geometric characteristics of fractures: the number of fractures (N_f), fracture length (L_f), fracture azimuth (θ_f), Inter-Branch Spacing (IBF), and deviation rate (R_d). Additionally, random sampling is applied to the L_f , θ_f , and R_d parameters to generate fractures with varying morphologies, enhancing the model's adaptability to different fracture configurations. In this study, random sampling was also conducted for matrix permeability (K_m),

fracture permeability (K_f), compressibility factor (C_i), bottom-hole pressure (P_{wf}), and bottom-hole pressure decline coefficient (α_{wf}). The random sampling was based on normal or uniform distributions with varying standard deviations and means. Table 1 and 2 summarize the basic physical properties and ranges of the random parameters for the reservoir and fluid in this study. Fig. 5 illustrates a randomly generated 3D reservoir model, with a hydraulically fractured horizontal well at the center and 10 randomly generated fractures of different shapes surrounding it.

Table 1—Physical properties of the reservoir and the fluid

Property	Value	Property	Value
Matrix porosity	0.1	Fracture_width	0.01 m
Fracture interval	80 m	Fracture porosity	0.4
Fracture porosity	0.40	Fracture compressibility	0.001
Thickness	30m	NTG	1
Oil volume factor	$3.02 \times 10^{-3} \text{MPa}^{-1}$	Oil phase viscosity	2 mPa·s
Secondary fracture permeability	10000 mD	Main fracture opening	10 mm
Secondary fracture opening	5 mm	Matrix rock compressibility	$1.0 \times 10^{-4} \text{Mpa}^{-1}$
Fracture rock compressibility	$1.0 \times 10^{-3} \text{MPa}^{-1}$	Water phase compressibility	$5 \times 10^{-4} \text{MPa}^{-1}$
Formation water volume coefficient	1.00001	Water phase viscosity	0.6 mPa·s
Initial reservoir pressure	20 MPa	Initial water saturation	0.15
Petroleum density(standard)	820kg/m ³	Formation water density(standard)	1000kg/m ³

Table 2—Random sampling parameters of reservoir and fluid

Property	Value range
Matrix permeability	$[10^{-8}, 10^{-3}]$ D
Fracture permeability	$[10, 40]$ D
Compressibility	$[10^{-4.1}, 10^{-2.9}]$ Mpa ⁻¹
Initial_WBHP	$[10, 20]$ Mpa
WBHP_K	$[0, 0.015]$ MPa/Day
Time	{1, 10, 50, 150, 300, 500} Day
Fracture length	$[0, 900]$ m
Deviation ratio	$[1, e^2]$
Fracture Azimuth	$[1/5\pi, 4/5\pi]$

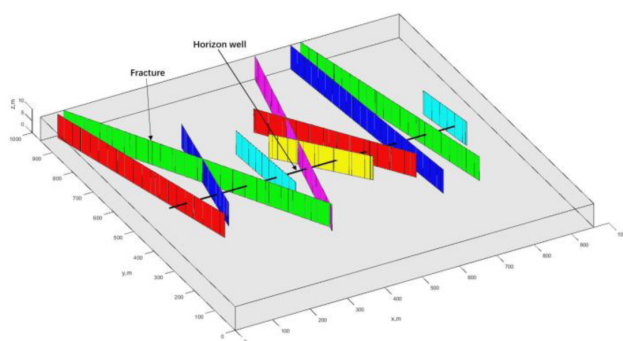


Figure 5—Schematic diagram of a three-dimensional geological model

Proxy model training and evaluation

This paper divides the dataset generated by EDFM into training and test sets. Each data sample consists of images of crack morphology, pressure distribution under different conditions and the corresponding constraint conditions. The training process involves setting a certain number of iterations, initializing the weights of the generator and critic models, and inputting the training set into the model in batches (batch size) until the entire dataset has been traversed. During training, the generator's input comprises crack morphology and the associated constraint conditions, while its output comprises pressure distribution data corresponding to the constraint conditions. The critic's input is the pressure distribution data and the corresponding constraint conditions. Its output is the distance value between the true sample distribution and the generator distribution. It is worth noting that the critic must be trained separately several times at each iteration to better learn and adapt to the differences between the real and generated data distributions, thereby providing more accurate feedback to the generator. Subsequently, the loss function propagates the error backwards and the Adam optimizer updates the generator and critic parameters. Through continuous iterative training, a model that meets the error requirements is ultimately obtained. The complete CWGAN-GP algorithm flow is shown in the [table 3](#):

Table 3—CWGAN-GP algorithm workflow

Algorithm 1 Condition WGAN with gradient penalty. We use default values of $\lambda=10$, $n_{critic}=5$, $\alpha=0.0002$, $\beta_1=0.5$, $\beta_2=0.999$.

Require: The gradient penalty coefficient λ , the number of critic iterations per generator iteration n_{critic} , the batch size m , Adam hyperparameters α, β_1, β_2 .

Require: initial critic parameters ω_0 , initial generator parameters θ_0 .

```

1: while  $\theta$  has not converged do
2:   for  $t = 1, \dots, n_{critic}$  do
3:     for  $i = 1, \dots, m$  do
4:       Sample real data  $x_f \sim p_r$ , fractures Shape  $x_f \sim p_g$ , a random number  $\epsilon \sim U[0, 1]$ , conditions  $c$ .
5:        $\tilde{x} \leftarrow G_\theta(x_f)$ 
6:        $\hat{x} \leftarrow \epsilon x + (1 - \epsilon)\tilde{x}$ 
7:        $L^{(i)} \leftarrow D_\omega(\hat{x}|c) - D_\omega(x_f|c) + \lambda \left( \nabla_{\hat{x}} D_\omega(\hat{x}|c)_2 - 1 \right)^2$ 
8:     end for
9:      $\omega \leftarrow \text{Adam}(\nabla_\omega \frac{1}{m} \sum_{i=1}^m L^{(i)}, \omega, \alpha, \beta_1, \beta_2)$ 
10:   end for
11:   Sample a batch of latent variables  $\{x_{f(i)}\}_{i=1}^m \sim p_g$ .
12:    $\theta \leftarrow \text{Adam}(\nabla_\theta \frac{1}{m} \sum_{i=1}^m D_\omega(G_\theta(x_{f(i)})), \theta, \alpha, \beta_1, \beta_2)$ 
13: end while

```

Performance metrics

Body text 2 paragraphs. During the training process, the Mean Squared Error (MSE) loss function is employed to quantitatively evaluate the discrepancy between the model's predictions and truth targets. The MSE metric is formally defined as:

$$Loss_{MSE}(y_i, \hat{f}(x_i)) = 1 - \frac{\sum_{i=1}^n (y_i - \hat{f}(x_i))^2}{n} \quad (5)$$

Therefore, the training process of the neural network model fundamentally constitutes an optimization procedure aimed at minimizing the mean squared error (MSE) loss function through iterative adjustment of model parameters.

During the testing phase, the predictive performance of the proposed surrogate model was rigorously evaluated using two quantitative metrics: the coefficient of determination (R^2) and root mean square error (RMSE), defined respectively as:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{f}(x_i))^2}{\sum_i (y_i - \bar{y})^2} \quad (6)$$

$$RMSE = \sqrt{E[(y_i - \hat{f}(x_i))^2]} \quad (7)$$

The dataset comprises 400 training cases, each containing pressure snapshots at six temporal intervals (totaling 2,400 pressure distributions). These were partitioned into training and validation sets at a 9:1 ratio. The model underwent 200 training epochs, with each epoch processing 5 randomly sampled cases until full traversal of the training data. Predictive performance was evaluated against EDFM-generated pressure fields through progressive monitoring of RMSE and R^2 metrics on both training and validation sets, as depicted in Figures 6,7. As the number of training iterations increases, the predictive accuracy of the neural network model exhibits rapid improvement followed by asymptotic convergence to stability. The testing set, rigorously excluded from the training process, serves to evaluate the model's generalization capability by assessing its predictive performance on out-of-sample data instances beyond the original training distribution. The close alignment between test set predictions and training set performance, as evidenced in the figures, demonstrates the model's robust generalization capability. Quantitative evaluation reveals exceptional predictive accuracy, with the test set achieving a coefficient of determination $R^2=0.995$ and root mean square error $RMSE<0.005$, thereby validating the CWGAN-GP framework's superior precision in pressure field estimation.

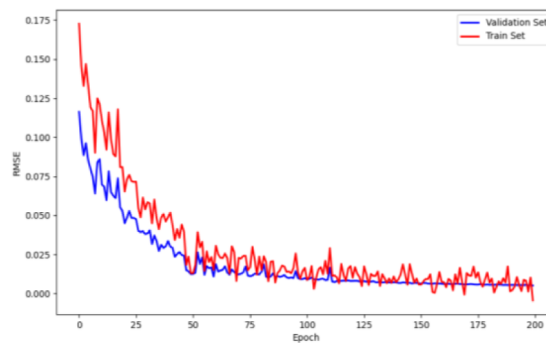


Figure 6—Diagram of RMSE during model training

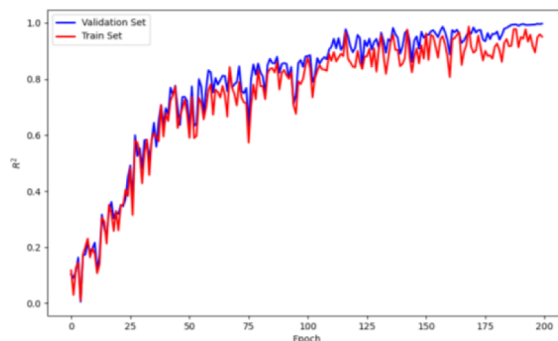


Figure 7—Diagram of R^2 during model training

Using fracture morphologies and constraint conditions beyond the training dataset as inputs, we conducted comparative pressure field predictions at multiple temporal intervals through both deep learning (CWGAN-GP) and EDFM approaches. The framework was rigorously evaluated across reservoirs with contrasting permeability characteristics: ultra-low permeability (0.15 mD), low-permeability (2.48 mD), and high-permeability (214.58 mD) systems, as demonstrated in Figures 8-10. We can find deep learning predictions exhibit congruent spatial-temporal evolution patterns with EDFM benchmarks. Pressure front propagation errors: <2.5% in high-permeability, <1.8% in ultra-low permeability cases.

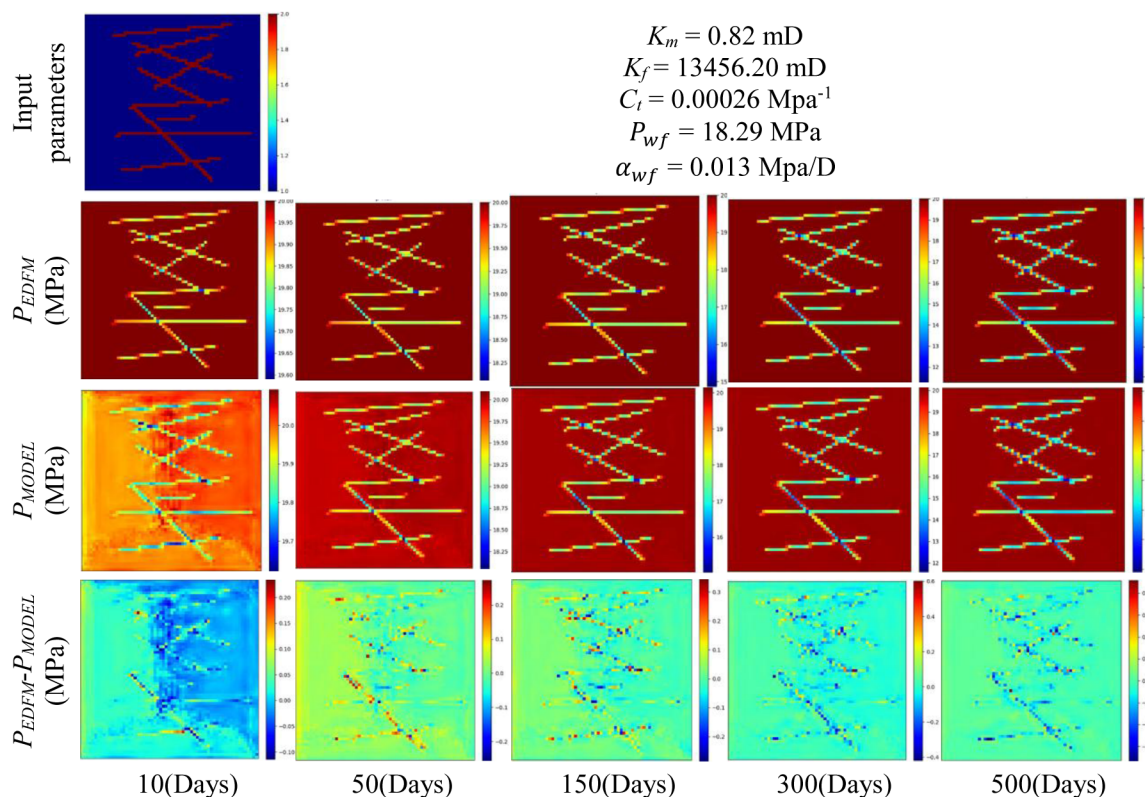


Figure 8—Input parameters includes fracture morphology, K_m , K_f , C_f , P_{wf} , α_{wf} (first row), pressure maps from the EDFM (second row) and CWGAN-GP proxy model (third row), along with difference maps (bottom row), at five timesteps (ultra-low permeability).

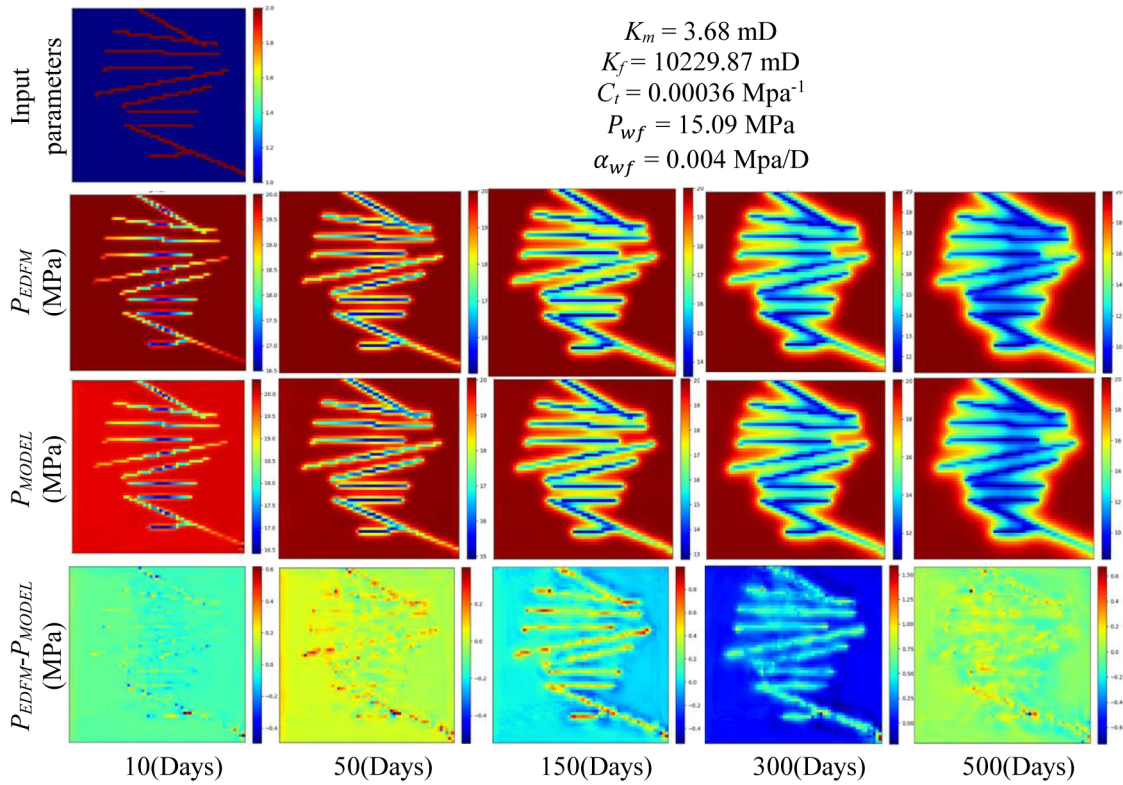


Figure 9—Input parameters includes fracture morphology, K_m , K_f , C_i , P_{wf} , α_{wf} (first row), pressure maps from the EDFM (second row) and CWGAN-GP proxy model (third row), along with difference maps (bottom row), at five timesteps (low-permeability).

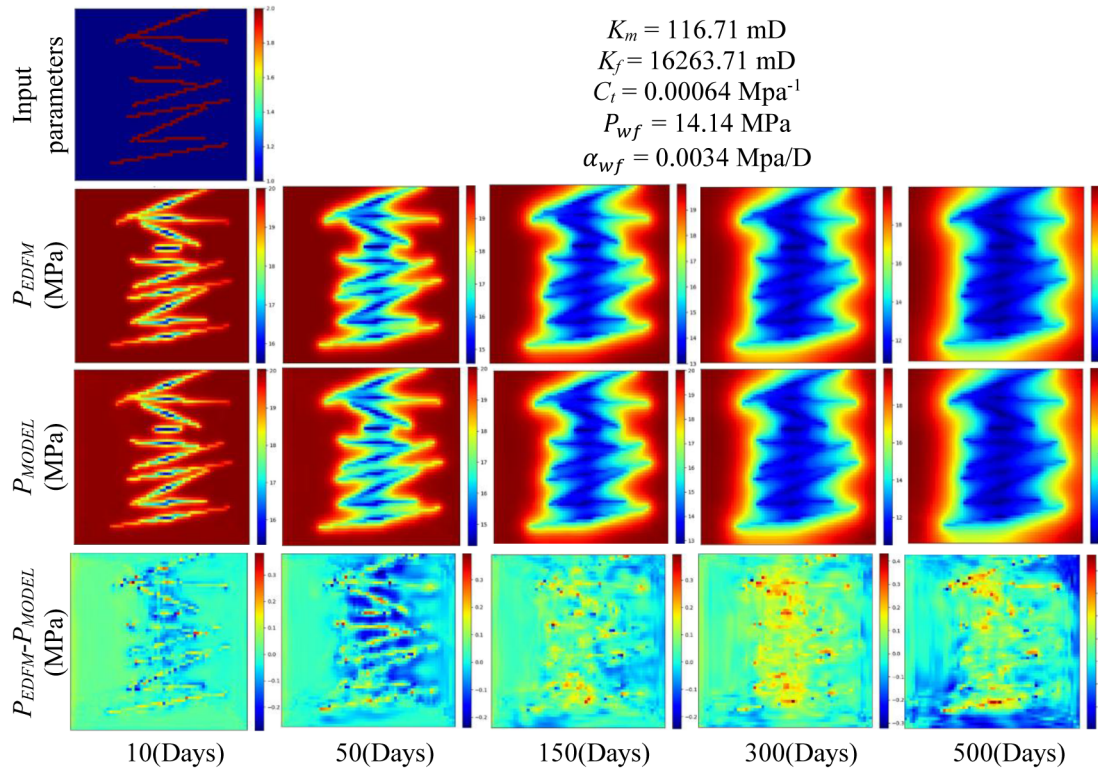


Figure 10—Input parameters includes fracture morphology, K_m , K_f , C_i , P_{wf} , α_{wf} (first row), pressure maps from the EDFM (second row) and CWGAN-GP proxy model (third row), along with difference maps (bottom row), at five timesteps (high-permeability).

The developed deep learning model demonstrates robust predictive capabilities for reservoir pressure distributions at arbitrary temporal intervals beyond the training dataset. The original training samples encompass five discrete time points for remaining oil saturation distribution: 10, 50, 150, 300, and 500 days. To validate extrapolation performance, five additional time points (30, 80, 100, 200, and 400 days) were randomly selected for pressure prediction using the trained model. Comparative results against numerical simulations (Fig. 11) reveal marginally larger discrepancies in pressure front propagation compared to in-sample time points, yet maintain prediction accuracy exceeding 90%. The predictive model demonstrates capability for instantaneous pressure estimation at arbitrary temporal intervals, with the trained neural network achieving exceptional computational efficiency—requiring merely 1.52 seconds to complete a single prediction on the specified hardware configuration (AMD EPYC 7642 48-core CPU, 256GB RAM, NVIDIA RTX A6000 GPU).

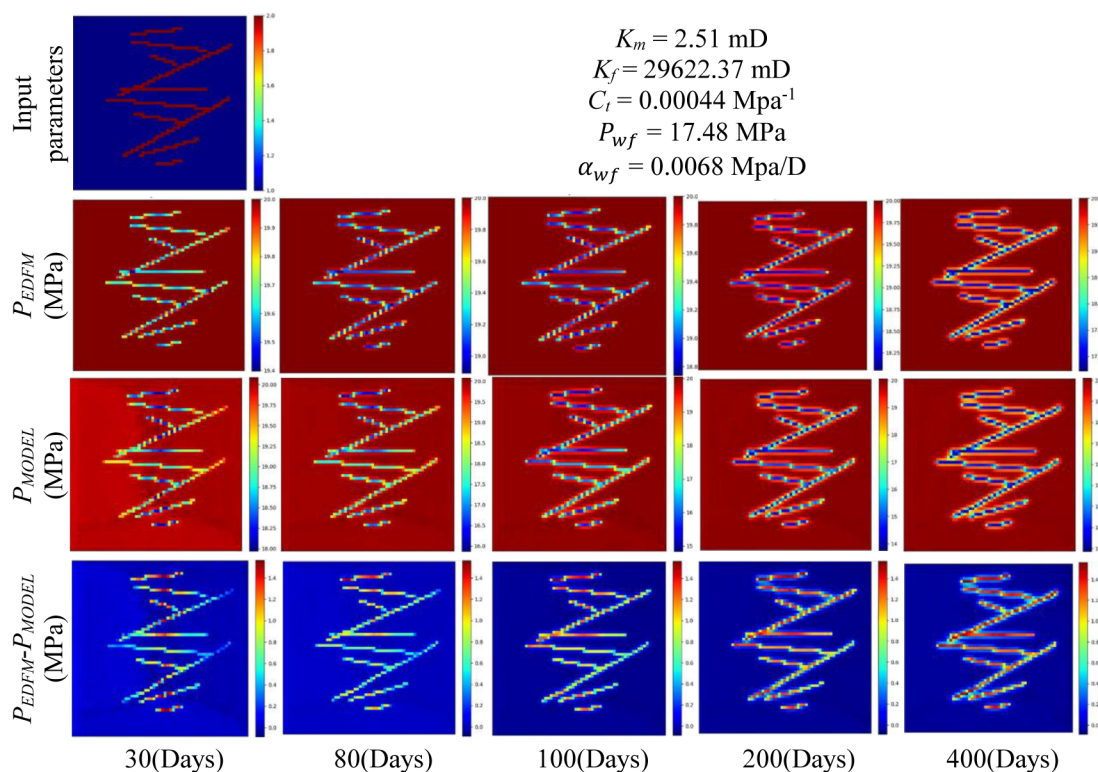


Figure 11—Input parameters includes fracture morphology, K_m , K_f , C_i , P_{wf} , α_{wf} (first row), pressure maps from the EDFM (second row) and CWGAN-GP proxy model (third row), along with difference maps (bottom row), at different five timesteps.

Although the deep learning model necessitates substantial simulated data for training dataset construction, its exceptional generalization capabilities ensure broad applicability across diverse reservoir scenarios. The trained surrogate model can be directly deployed for pressure prediction under arbitrary conditions, provided that the input parameters fall within the coverage of the training dataset's parameter space. Furthermore, when the training dataset encompasses an extensive range of geological and constraints, incorporates enough input-output mappings, and is supported by advanced computational infrastructure, our framework demonstrates foundational potential for developing large-scale intelligent prediction systems capable of pan-reservoir dynamic forecasting. This scalability potential forms the foundation for our ongoing research into developing next-generation foundation models for reservoir management.

Conclusions

In this study, we developed an efficient and practical framework for reservoir pressure prediction by CWGAN-GP proxy model. The CWGAN-GP architecture synergizes the strengths of CGAN and WGAN, ensuring robust convergence under multi-parametric input conditions while significantly enhancing model applicability and training stability. Our trained CWGAN-GP surrogate model achieves high-fidelity pressure predictions under arbitrary operational and geological constraints. Based on these findings, we derive the following conclusions:

1. The CWGAN-GP exhibits consistent predictive accuracy on both training ($R^2=0.997$) and test sets ($R^2=0.995$), with a generalization gap of <2% RMSE.
2. The framework enables simultaneous pressure prediction across arbitrary temporal, well-control, and geological conditions, achieving 92.3% accuracy in multi-constrained scenarios.
3. The hybrid loss function eliminates convergence issues prevalent in traditional deep learning models, suppressing gradient vanishing/explosion through Lipschitz continuity enforcement.
4. The proxy model reduces simulation time by 3 orders of magnitude, demonstrating potential for: history matching acceleration; production optimization; foundation model development for unconventional reservoir dynamic prediction's large data model.

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